Linear Arboricity

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The **linear arboricity** of a graph G, denoted by $la(G)$, is the minimum number of linear forests into which the edge set $E(G)$ can be partitioned.

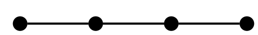
Equivalently, we seek to colour every edge of G using as few colours as possible so that every colour class induces a linear forest.

Intuition A linear forest is simply a collection of disjoint paths. Therefore, the linear arboricity asks the following question:

> How many collections of paths are needed so that every edge of the graph belongs to exactly one collection?

The smaller this number, the "closer" the graph is to already being a collection of paths.

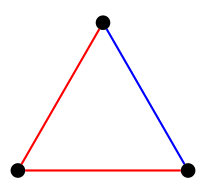
Example 1: A Path Consider the path P_4.



The graph itself is already a linear forest since it consists of exactly one path.
Hence,

$$la(P_4) = 1.$$

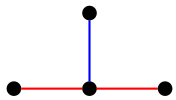
Example 2: A Triangle Consider the cycle C_3.



The red edges form one path, while the blue edge forms another path.
Each colour therefore represents one linear forest.
Hence,

$$la(C_3) = 2.$$

Example 3: A Star Consider the star graph $K_{1,3}$.



Although $K_{1,3}$ is a tree, it is not a linear forest because the centre vertex has degree 3.
The red edges form one path and the blue edge forms another.
Therefore,

$$la(K_{1,3}) = 2.$$

Remark Notice that the graph itself does *not* have to be a linear forest. We only require that its edges can be partitioned into linear forests.

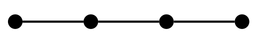
Linear Forest

A **linear forest** is a forest in which every connected component is a path.
Equivalently, a graph is a linear forest if and only if

- it contains no cycles, and
- every vertex has degree at most 2.

The first condition prevents cycles, while the second prevents branching.

Examples The following are linear forests.

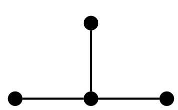


One path

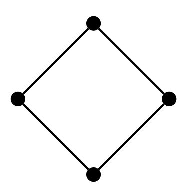
Two disjoint paths

Both graphs consist entirely of paths.
Hence both are linear forests.

Non-examples



Branching



Cycle

The graph on the left is a tree but not a path because one vertex has degree 3.
The graph on the right contains a cycle.
Therefore, neither graph is a linear forest.

Remarks

- Every path is a linear forest.
- Every linear forest is a forest.
- Every connected linear forest is a path.
- Every vertex of a linear forest has degree at most 2.

Forest

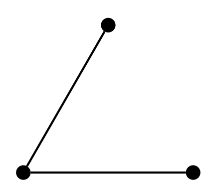
A **forest** is a graph that contains no cycles.

Equivalently, every connected component of a forest is a tree.

Examples

This graph is a forest consisting of two trees.
A connected forest is simply called a tree.

Non-example



This graph is not a forest because it contains a cycle.

Remarks

- Every tree is a forest.
- A forest may be connected or disconnected.
- A forest with exactly one connected component is a tree.

$$\boxed{\text{Tree} \subseteq \text{Forest.}}$$